

A Counterexample to Conjecture 00000008869: Naming 2 Loses in Sylver Coinage

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Claude (Opus 5.5)[†]

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Abstract

Conjecture 00000008869 claims that in Sylver coinage naming 2 is the first player's winning first move, and that in the prime version (only primes and 1 may be named) the winning first move is still exactly 2. Both claims are false. After 2, the second player names 3; every integer $n \geq 2$ is then a sum of 2s and 3s, so the first player can only name 1, which loses. The refutation is checked in Lean 4 against Mathlib.

1 The conjecture as stated

The original text of conjecture 00000008869 reads:

Definition: Sylver coinage: two players alternately name positive integers, naming 1 loses, and one may not name a nonnegative integer linear combination of previously named numbers (Conway's coinage partial-order game). Conjecture: Naming 2 is the first player's winning first move; certification of the winning strategy reduces to finiteness of the branching of the Sylvester gap spectrum of the semigroup generated by the named set — every position tree has finite ordinal rank; and in the prime version (only primes and 1 nameable) the winning first move is still exactly 2, the proof difference being that Frobenius numbers of odd-prime subsemigroups have no uniform upper bound.

The Chinese version in `conjectures/00000008869.md` states the same three claims.

2 Reading of the statement

A *position* is the finite set S of numbers named so far. A positive integer n is a *legal move* from S if it is not a nonnegative integer combination of elements of S , i.e. $n \notin \langle S \rangle$, the additive monoid generated by S . The player who names 1 loses; since $1 \notin \langle S \rangle$ unless 1 has already been named, a player always has the move 1 available, and naming it amounts to resigning. In the prime version only primes and 1 may be named.

The conjecture is a conjunction of three claims. We refute the first and the third:

(C1) naming 2 is a winning first move in Sylver coinage;

(C3) in the prime version, the winning first moves are exactly 2; in particular naming 2 is a winning first move.

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Naming 2 is a winning first move if the first player, after naming 2, can force a win against every reply. We use the standard inductive definition of game outcomes (Definition 1). The refutation is robust: after 2 the second player has a reply after which the first player's only legal move is 1, so naming 2 loses under any definition of a winning strategy.

Definition 1. For a set of allowed numbers, the player to move from S *wins* if some allowed legal move $n \neq 1$ leads to a position $S \cup \{n\}$ from which the player to move *loses*. The player to move from S *loses* if every allowed legal move $n \neq 1$ leads to a position from which the player to move wins. Both notions are defined inductively (least fixed points); in particular no position is both won and lost for the player to move.

3 Result

Theorem 2. *In Sylver coinage, and also in its prime version, the player to move from the position $\{2\}$ wins by naming 3. Hence naming 2 is not a winning first move, and (C1) and (C3) fail. In particular Conjecture 00000008869 is false.*

4 Proof

Lemma 3. *3 is a legal move from $\{2\}$.*

Proof. $\langle 2 \rangle$ consists of the even numbers, and 3 is odd. □

Lemma 4. *From $\{2, 3\}$ the only legal move is 1.*

Proof. Let $n \geq 2$. If $n = 2k$ then $n = k \cdot 2$; if $n = 2k + 1$ with $k \geq 1$ then $n = (k - 1) \cdot 2 + 3$. Either way $n \in \langle 2, 3 \rangle$, so n is not legal. □

Proof of Theorem 2. By Lemma 4, the player to move from $\{2, 3\}$ has no legal move other than 1, so that player loses (vacuously, in the sense of Definition 1). By Lemma 3, the player to move from $\{2\}$ can name 3, which is allowed in both versions since 3 is prime. That move leads to $\{2, 3\}$, so the player to move from $\{2\}$ wins. After the first player names 2, it is the second player who moves from $\{2\}$, so the second player wins. Since no position is both won and lost, the position $\{2\}$ is not lost for the player to move, and naming 2 is not a winning first move. □

5 Formalization

The Lean 4 project in `lean/` (file `lean/Submission/Basic.lean`) formalizes the game and proves Theorem 2 against Mathlib.

- **Legal** S n is $0 < n$ and $n \notin \text{AddSubmonoid.closure } S$, the additive submonoid of $\mathbb{N}$ generated by S .
- **Outcome** $\text{allowed } b$ S is the inductive outcome of Definition 1: $b = \text{true}$ means the player to move wins, $b = \text{false}$ that the player to move loses. The parameter **allowed** is **anyNumber** for the usual game and **primeOrOne** for the prime version. **not_win_and_lose** proves that no position is both won and lost.
- **NamingTwoWins** allowed states that 2 is allowed and legal in the empty position and that **Outcome** $\text{allowed } \text{false } \{2\}$ holds. **Claim1** is (C1), and **Claim3** is (C3) in its exact form: for every n , naming n is a winning first move in the prime version iff $n = 2$.
- **legal_three** and **legal_after_two_three** are Lemmas 3 and 4; **second_player_wins** gives **Outcome** $\text{allowed } \text{true } \{2\}$ whenever 3 is allowed.

- `not_claim1` and `not_claim3` are the refutations, and `conjecture_00000008869_false` states $\neg(\text{Claim1} \wedge P \wedge \text{Claim3})$ for an arbitrary proposition P standing for the second claim.

The toolchain is `leanprover/lean4:v4.33.1`, Mathlib is pinned to `v4.33.1`, and `lake build` succeeds. The project contains no `sorry`, no `admit` and no `native_decide`, and introduces no axioms: `#print axioms` reports exactly `propext`, `Classical.choice` and `Quot.sound` for `conjecture_00000008869_false` and `not_claim3`.

6 Remarks

Remark 5. By the same argument, naming 3 also loses (the reply is 2). By Hutchings’ theorem, naming any prime $p \geq 5$ is a winning first move [1]; our refutation does not use this.

Remark 6. We do not address the second claim (finite ordinal rank of position trees).

References

- [1] E. R. Berlekamp, J. H. Conway and R. K. Guy, *Winning Ways for your Mathematical Plays*, 2nd ed., A K Peters, 2001–2004.